

$C_{\phi d}^{12} i_1 i_2 \rightarrow$

$$\begin{aligned} & \frac{1}{16\pi^2} \left(-\frac{1}{9} \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pr}} a_d^{pr} \text{LF}_{2,2,0} [m_d^r, m_q^p] \delta_{i_1 i_2} + \frac{1}{18} \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pr}} a_d^{pr} \text{LF}_{3,2,-1} [m_d^r, m_q^p] \delta_{i_1 i_2} - \right. \\ & \frac{1}{9} \tilde{\mu} \mathbf{g}_1^2 \overline{y_e^{pr}} a_e^{pr} \text{LF}_{2,2,0} [m_e^r, m_l^p] \delta_{i_1 i_2} + \frac{1}{18} \tilde{\mu} \mathbf{g}_1^2 \overline{y_e^{pr}} a_e^{pr} \text{LF}_{3,2,-1} [m_e^r, m_l^p] \delta_{i_1 i_2} + \\ & \frac{1}{9} \tilde{\mu} \mathbf{g}_1^2 \overline{y_e^{pr}} a_e^{pr} \text{LF}_{3,1,0} [m_l^p, m_e^r] \delta_{i_1 i_2} + \frac{1}{9} \tilde{\mu} \mathbf{g}_1^2 \overline{y_e^{pr}} a_e^{pr} \text{LF}_{3,2,-1} [m_l^p, m_e^r] \delta_{i_1 i_2} - \\ & \frac{1}{4} \tilde{\mu} \mathbf{g}_1^2 \overline{y_e^{pr}} a_e^{pr} \text{LF}_{4,1,-1} [m_l^p, m_e^r] \delta_{i_1 i_2} + \frac{1}{9} \tilde{\mu} \mathbf{g}_1^2 \overline{y_e^{pr}} a_e^{pr} \text{LF}_{5,1,-2} [m_l^p, m_e^r] \delta_{i_1 i_2} + \\ & \frac{1}{9} \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pr}} a_d^{pr} \text{LF}_{3,1,0} [m_q^p, m_d^r] \delta_{i_1 i_2} + \frac{1}{9} \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pr}} a_d^{pr} \text{LF}_{3,2,-1} [m_q^p, m_d^r] \delta_{i_1 i_2} - \\ & \frac{5}{12} \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pr}} a_d^{pr} \text{LF}_{4,1,-1} [m_q^p, m_d^r] \delta_{i_1 i_2} + \frac{1}{3} \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pr}} a_d^{pr} \text{LF}_{5,1,-2} [m_q^p, m_d^r] \delta_{i_1 i_2} + \\ & \frac{1}{18} \tilde{\mu} \mathbf{g}_1^2 \overline{y_u^{pr}} a_u^{pr} \text{LF}_{2,2,0} [m_q^p, m_u^r] \delta_{i_1 i_2} - \frac{1}{36} \tilde{\mu} \mathbf{g}_1^2 \overline{y_u^{pr}} a_u^{pr} \text{LF}_{3,2,-1} [m_q^p, m_u^r] \delta_{i_1 i_2} - \\ & \frac{1}{18} \tilde{\mu} \mathbf{g}_1^2 \overline{y_u^{pr}} a_u^{pr} \text{LF}_{3,1,0} [m_u^r, m_q^p] \delta_{i_1 i_2} - \frac{1}{18} \tilde{\mu} \mathbf{g}_1^2 \overline{y_u^{pr}} a_u^{pr} \text{LF}_{3,2,-1} [m_u^r, m_q^p] \delta_{i_1 i_2} - \\ & \frac{1}{6} \tilde{\mu} \mathbf{g}_1^2 \overline{y_u^{pr}} a_u^{pr} \text{LF}_{4,1,-1} [m_u^r, m_q^p] \delta_{i_1 i_2} + \frac{1}{3} \tilde{\mu} \mathbf{g}_1^2 \overline{y_u^{pr}} a_u^{pr} \text{LF}_{5,1,-2} [m_u^r, m_q^p] \delta_{i_1 i_2} - \\ & \frac{1}{12} m_1 \tilde{\mu} \mathbf{g}_1^4 \text{LF}_{4,1,-1} [\tilde{\mu}, m_1] \delta_{i_1 i_2} + \frac{1}{9} m_1 \tilde{\mu} \mathbf{g}_1^4 \text{LF}_{5,1,-2} [\tilde{\mu}, m_1] \delta_{i_1 i_2} - \\ & \frac{1}{4} m_2 \tilde{\mu} \mathbf{g}_1^2 \mathbf{g}_2^2 \text{LF}_{4,1,-1} [\tilde{\mu}, m_2] \delta_{i_1 i_2} + \frac{1}{3} m_2 \tilde{\mu} \mathbf{g}_1^2 \mathbf{g}_2^2 \text{LF}_{5,1,-2} [\tilde{\mu}, m_2] \delta_{i_1 i_2} - \\ & \frac{1}{36} m_1 \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{2,1,1,0} [m_1, m_d^{i1}, m_q^p] + \frac{1}{72} m_1 \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} y_d^{pi2} \\ & \text{LF}_{2,2,1,-1} [m_1, m_d^{i1}, m_q^p] + \frac{1}{36} m_1 \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{3,1,1,-1} [m_1, m_d^{i1}, m_q^p] - \\ & \frac{1}{72} m_1 \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{2,2,1,-1} [m_1, m_q^p, m_d^{i1}] - \frac{1}{12} m_1 \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{2,1,1,0} [m_1, m_q^p, \tilde{\mu}] + \\ & \frac{1}{12} m_1 \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{3,1,1,-1} [m_1, m_q^p, \tilde{\mu}] + \frac{1}{12} m_1 \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_d^{i1}] + \\ & \frac{1}{6} m_1 \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_q^p] - \frac{3}{4} m_2 \tilde{\mu} \mathbf{g}_2^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{2,1,1,0} [m_2, m_q^p, \tilde{\mu}] + \\ & \frac{3}{4} m_2 \tilde{\mu} \mathbf{g}_2^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{3,1,1,-1} [m_2, m_q^p, \tilde{\mu}] + \frac{3}{4} m_2 \tilde{\mu} \mathbf{g}_2^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_q^p] + \\ & \frac{2}{3} m_3 \tilde{\mu} \mathbf{g}_3^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{2,1,1,0} [m_3, m_d^{i1}, m_q^p] - \frac{1}{3} m_3 \tilde{\mu} \mathbf{g}_3^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{2,2,1,-1} [m_3, m_d^{i1}, m_q^p] - \\ & \frac{2}{3} m_3 \tilde{\mu} \mathbf{g}_3^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{3,1,1,-1} [m_3, m_d^{i1}, m_q^p] + \frac{1}{3} m_3 \tilde{\mu} \mathbf{g}_3^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{2,2,1,-1} [m_3, m_q^p, m_d^{i1}] - \\ & \frac{1}{8} \tilde{\mu} \overline{y_d^{pi1}} y_d^{ri2} \overline{y_u^{rs}} a_u^{ps} \text{LF}_{2,2,1,-1} [m_q^p, \tilde{\mu}, m_u^s] + \\ & \frac{1}{8} \tilde{\mu} \overline{y_d^{pi1}} y_d^{ri2} \overline{y_u^{rs}} a_u^{ps} \text{LF}_{2,2,1,-1} [m_u^s, \tilde{\mu}, m_q^p] - \frac{1}{6} m_1 \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_1, m_d^{i1}] + \\ & \frac{1}{6} m_1 \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_1, m_d^{i1}] + \frac{1}{4} \tilde{\mu} \overline{y_d^{pi1}} y_d^{ri2} \overline{y_u^{rs}} a_u^{ps} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_q^p, m_u^s] - \\ & \frac{1}{4} \tilde{\mu} \overline{y_d^{pi1}} y_d^{ri2} \overline{y_u^{rs}} a_u^{ps} \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_q^p, m_u^s] + \frac{1}{12} m_1 \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} y_d^{pi2} \\ & \text{LF}_{1,1,1,1,0} [m_1, m_d^{i1}, m_q^p, \tilde{\mu}] + \frac{1}{24} m_1 \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{2,1,1,1,-1} [m_1, m_d^{i1}, m_q^p, \tilde{\mu}] + \\ & \frac{1}{12} \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} a_d^{pi2} \text{LF}_{1,1,1,1,0} [m_1, m_d^{i2}, m_q^p, \tilde{\mu}] - \frac{1}{24} \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} a_d^{pi2} \\ & \text{LF}_{2,1,1,1,-1} [m_1, m_d^{i2}, m_q^p, \tilde{\mu}] - \frac{1}{4} \tilde{\mu} \overline{y_d^{pi1}} \overline{y_d^{rs}} y_d^{ri2} a_d^{ps} \text{LF}_{2,1,1,1,-1} [m_d^s, m_q^p, m_q^r, \tilde{\mu}] + \\ & \frac{1}{24} m_1 \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{2,1,1,1,-1} [m_d^{i1}, m_1, m_q^p, \tilde{\mu}] - \frac{1}{24} \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} a_d^{pi2} \\ & \text{LF}_{2,1,1,1,-1} [m_d^{i2}, m_1, m_q^p, \tilde{\mu}] + \frac{1}{18} \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} a_d^{pi2} \text{LF}_{2,1,1,1,-1} [m_q^p, m_1, m_d^{i1}, m_d^{i2}] - \\ & \frac{1}{24} m_1 \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} y_d^{pi2} \text{LF}_{2,1,1,1,-1} [m_q^p, m_1, m_d^{i1}, \tilde{\mu}] + \frac{1}{24} \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} a_d^{pi2} \\ & \text{LF}_{2,1,1,1,-1} [m_q^p, m_1, m_d^{i2}, \tilde{\mu}] + \frac{2}{3} \tilde{\mu} \mathbf{g}_3^2 \overline{y_d^{pi1}} a_d^{pi2} \text{LF}_{2,1,1,1,-1} [m_q^p, m_3, m_d^{i1}, m_d^{i2}] + \\ & \frac{1}{4} \tilde{\mu} \overline{y_d^{pi1}} y_d^{ri2} \overline{y_u^{rs}} a_u^{ps} \text{LF}_{2,1,1,1,-1} [m_u^s, m_q^p, m_q^r, \tilde{\mu}] - \frac{1}{24} m_1 \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} y_d^{pi2} \\ & \left. \text{LF}_{2,1,1,1,-1} [\tilde{\mu}, m_1, m_d^{i1}, m_q^p] + \frac{1}{24} \tilde{\mu} \mathbf{g}_1^2 \overline{y_d^{pi1}} a_d^{pi2} \text{LF}_{2,1,1,1,-1} [\tilde{\mu}, m_1, m_d^{i2}, m_q^p] \right) \end{aligned}$$